Seminar Introduction

palmgroup

Institute of Meteorology and Climatology, Leibniz Universität Hannover

Seminar Introduction

Content

- Seminar goals
- PALM group at IMUK
- Lecturers & exercise staff
- Schedule & room plan
- Lunch & coffee breaks
- Get-together-events
- WiFi connections
- Additional remarks

Seminar Introduction

Seminar goals

- Get to know & work with PALM*

  * formerly abbreviation for "PArallelized Large-eddy simulation Model"

For this we provide:

- Basic knowledge about LES
- Introduction to the PALM code
  - Code structure
  - Numerics & Parallelization
  - Specific features, e.g. PALM's land surface model, topography model, building surface model, wind turbine model, chemistry model, etc.
- **Hands-on sessions**
  - PALM installation (already completed in advance!)
  - Run PALM for different application scenarios
  - Work with user-specified code
  - Understand & analyze PALM output data (and LES data in general)

Seminar Introduction

PALM group at IMUK

PALM group in front of the Castle Herrenhausen next to the Royal Gardens, July 2023.

About us https://www.meteo.uni-hannover.de/en/research/boundary-layer-meteorology

PALM website http://palm-model.org

Seminar Introduction

Lecturers (responsibilities, fields of expertise)

- **Prof. Siegfried (Siggi) Raasch (Leibniz University Hannover, LUH)**
  - Professor Emeritus, former head of Boundary-Layer Meteorology group at LUH
    (code supervisor, father of PALM)

- **Prof. Björn Maronga (LUH)**
  - Professor, current head of Boundary-Layer Meteorology group (urban and rural boundary layers, fog, near-surface processes, wind energy)

- **Dr. Micha Gryschka (LUH)**
  - Senior Scientist (lecturer at IMUK, cold-air outbreaks, flow over leads)

Seminar Introduction

Lecturers (responsibilities, fields of expertise)

- **Dr. Sebastian Schubert (TU Berlin)**
  - Senior Scientist (canopy/radiation modeling, static drivers)

- **Dr. Sabine Banzhaf (FU Berlin)**
  - Senior Scientist (air chemistry)

Seminar Introduction

Hands-on (Exercise) session staff

Additional exercise staff:

- **Sarah Stenzel** (PhD student, polar boundary layers)
- **Giovanna Motisi** (PhD student, traffic induced turbulence)
- **Jannis Brassat** (PhD student, wind energy)
- **Yasmin Ahlert** (PhD student, wind energy)
- **Nooshin Nowzamani** (PhD, urban boundary layers, buildings physics)
- **Pierre Lampe** (PhD student, urban boundary layers)

Advisors for exercises:

- Monday 15: Siggi, Björn, Sarah, Jannis
- Tuesday 16: Siggi, Sarah, Yasmin, Micha
- Wednesday 17: Siggi, Björn, Giovanna, Nooshin, Pierre
- Thursday 18: Siggi, Björn, Nooshin
- Friday 19: Siggi, Björn, Jannis, Pierre, Yasmin

or

Seminar Introduction

Local staff

Local staff for catering & Administration:

- Sarah Stenzel (organizer)
  - Master student (polar boundary layers)

- Bastian Blume & Mert Hanoglu (catering/coffee breaks)

- Berit Parbel & Petra Kraege (administration)

Seminar Introduction

Introduction of participants

- That's who we are …
- **… so, who are you? Please stand up and tell us your…**
  - Name
  - Institution/residence
  - Job position
  - Reason for your participation

Seminar Introduction

Schedule

Lectures | Room 009 | Hands-on | Room F118 & F139 & F128 | Ice Breaker | Room F128

	Monday	Tuesday	Wednesday	Thursday	Friday
8^{30}-10^{00} coffee break	Seminar Introduction & Fundamentals of LES Maronga/Raasch	Numerics & Boundary Conditions Raasch Code Parallelization Raasch	Restart Runs, Non-Cyclic Boundary Conditions, User-defined code Gryschka	Land Surface Model (LSM) & Urban Surface Model (USM) Maronga	Bulk Cloud Model (BCM) Gryschka
10^{30}-11^{45} short break	LES for Beginners & PALM Overview Raasch	Code Structure & Features Raasch Topography Raasch	Grid Nesting Raasch	Radiation Models Schubert Dynamic Drivers Maronga	Wind Turbine Model (WTM) **OR** *Chemistry (Room F118) Maronga / Banzhaf
12^{00}-13^{00}	PALM Steering Raasch Data Visualization Maronga	Debugging Raasch	Graphical User Interface: palmrungui Maronga	Static Drivers Schubert	E5: Wind Turbine Model E6: Chemistry (self-learning) Maronga
13^{00} -14^{30}	Lunch Break	Group Photo & Lunch Break	Lunch Break	Lunch Break	Lunch Break
14^{30}-15^{00}	E1: Convective Boundary Layer Maronga	E2: Neutral Boundary Layer Gryschka	E3: Flow Around Buildings Maronga	E4: Land Surface Model Maronga	E5/E6
15^{00}-16^{00} coffee break	E1	E2	E3	E4	Discussion of Results E5/Problems Maronga
16^{30}-17^{00}	E1	E2	E3	E4	Final remarks / Closing Raasch/Maronga
17^{00}-18^{00}	Discussion of Results/Problems Maronga	Discussion of Results/Problems Gryschka	Discussion of Results/Problems Maronga	Discussion of Results/Problems Maronga	
18^{00}-21^{00}	Ice Breaker		Seminar Dinner		

Updates of the schedule & exercise material:
https://seafile.cloud.uni-hannover.de/d/ed4a9c691a874d43966f/

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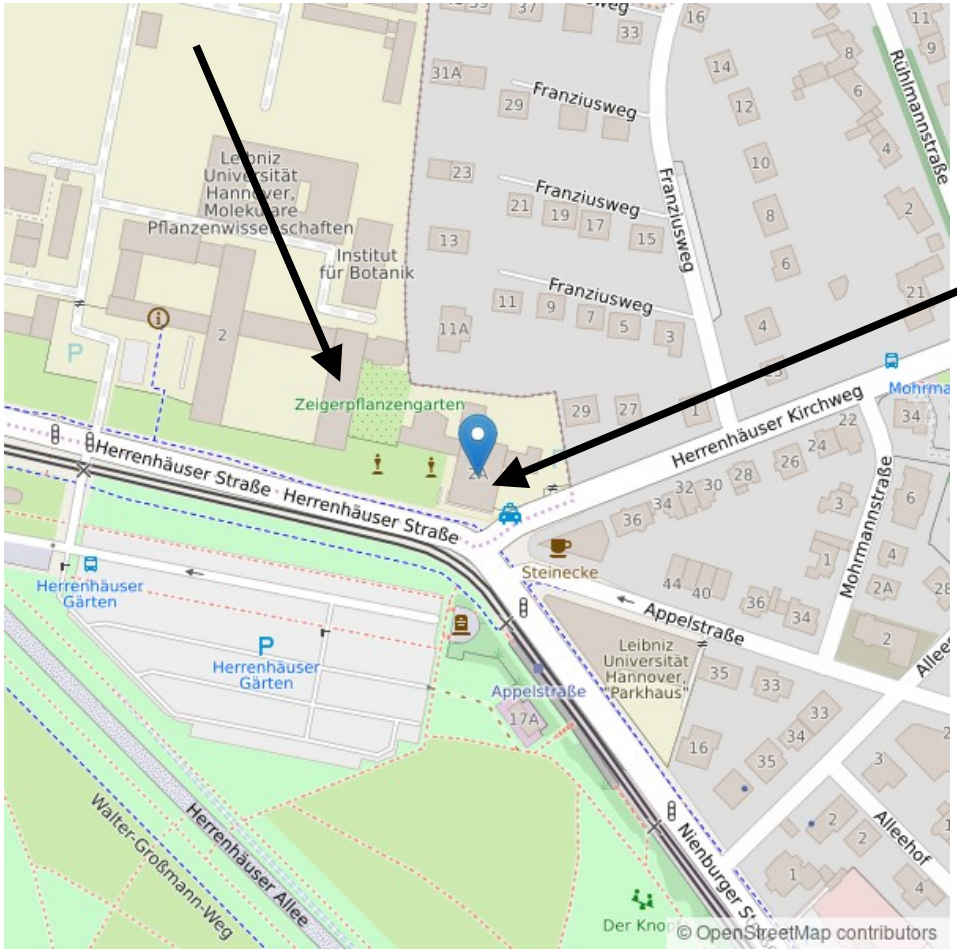
Exercise sessions

- PALM installation and exercises will be carried out on your private notebooks.
- In case of any technical problems with your private notebooks that can't be solved within a short time, we might ask you to work together with one of the other participants.
- You will find your assigned exercise room number that on your name tag.

Seminar Introduction

Location of rooms

Exercises, coffee breaks, ice breaker

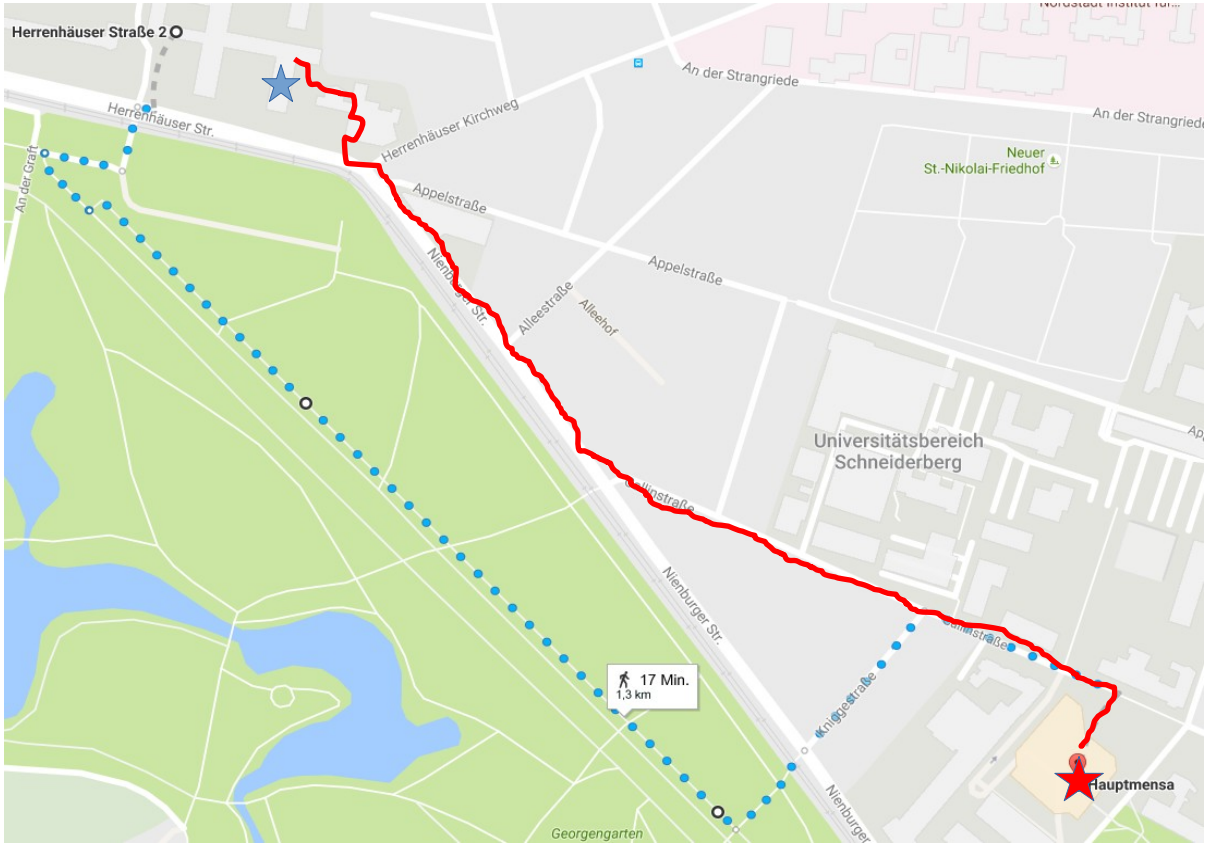


Lecture Hall

Seminar Introduction

Lunch & coffee breaks

- Lunch break (1:00 pm - 2:30 pm) e.g. at university canteen



shortest way

- Coffee breaks in the morning (10:00 am) and in the afternoon (4:00 pm). Coffee, tea, and snacks will be provided in the hallway of the institute.

Seminar Introduction

Ice breaker

- After today's lectures/exercise, we would like to invite you to a „get together" with drinks and finger food in the „Weather Room" (F139) plus hallway, starting from about 6 pm.

All drinks are alcohol-free!

Vegan / vegetarian / non-vegetarian finger food available!

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Optional dinner (Wednesday)

- Joint dinner (not included in seminar fee).
- @ Lieblingsbar (Herrenhäuser Markt 5), Wednesday 16, 6:30 p.m.
- Registration already completed. Late birds… please contact Sarah!

Seminar Introduction

WiFi connection

You can connect via:

- **eduroam** (if you already have your own personal account) or
- **eduroam** (with guest account, please sign list and **take ticket from Sarah!**)

Ausgabeliste der Zugangstickets "mukluh" Seite 1/3

Nutzungszeitraum: 10.02.2020-14.02.2020 Nutzungsdauer: 5 Tage

Benutzername	Vor- und Nachname	E-Mail-Adresse
mukluh001		
mukluh002		
mukluh003		
mukluh004		

Zugangsticket für das WLAN der Leibniz Universität Hannover

Benutzername: mukluh001@uni-hannover.de

Kennwort: hC5V6eb3

Netzwerkname (SSID): eduroam

Nutzungszeitraum: 10.02.2020-14.02.2020

Nutzungsdauer: 5 Tage (ab erstmaligem Login)

Mit der Nutzung akzeptieren Sie die Regelungen zur Nutzung der IT-Systeme und -Services des LUIS, nachzulesen auf unserer Website.

For access to the eduroam WLAN see also:
https://www.luis.uni-hannover.de/en/services/communication/netz/angebote-im-service-netz/wlan/wireless-lan-configuration

Technical parameters in short:

- Name(ESSID): eduroam
- Encryption: WPA2 (AES)
- Authentication: PEAP-MSCHAPv2
- Server certificate derived from root certificate "T-Telesec Global Root Class 2"

Seminar Introduction

Additional remarks

- During the lectures and exercises, feel free to ask questions as often and whenever you like!

- Seminar certificates will be provided on the last day of the seminar

Leibniz Universität Hannover

CERTIFICATE

OF PARTICIPATION

This certificate is presented to

in recognition of the active participation in the 5-day seminar

An Introduction

held on at Leibniz University Hannover, Germany.

The course content covered the fundamentals of Large-Eddy-Simulation and the components of the PALM model system as well as its application. The holder of this certificate has proven to have gained a deeper understanding of the LES technique and is able to apply PALM for different meteorological scenarios, including analysis of model data output.

The participation in this seminar is equivalent to PALM certificates LEVEL 1 and LEVEL 2.

DATE

SIGNATURE

Prof. Dr. Björn Maronga
Institute of Meteorology and Climatology
Leibniz University Hannover

PALM online:
http://palm-model.org

Our YouTube channel:
https://youtube.com/user/palmhannover

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